

Chapter 07 Rotational Motion and The Law of Gravity

The Radian

A unit of angular measure. Learned before.

Angular Displacement

Axis of rotation is the center of the disk. For a **fixed reference line**, the angle θ is the angular displacement during time t .

Rigid Body

- **Every point** on the object undergoes circular motion about the point O.
- **All parts** of the object of the body rotate through the **same angle during the same time**.
That is, **each part** of the body is **fixed** in position relative to all **other parts** of the body. And this object is called **rigid body**.

Average Angular Speed

The ratio of the angular displacement to the time interval

$$\omega_{av} = \frac{\Delta\theta}{\Delta t}$$

Average Angular Acceleration

The ratio of the angular speed to the time interval

$$\alpha_{av} = \frac{\Delta\omega}{\Delta t}$$

For a rigid body, every point has the same angular speed and acceleration.

Analogies Between Linear and Rotational Motion

Linear Motion with a Constant (Variables: x and v)	Rotational Motion about a Fixed Axis with α Constant (Variables: θ and ω)
$v = v_i + at$	$\omega = \omega_i + \alpha t$ [7.7]
$\Delta x = v_i t + \frac{1}{2}at^2$	$\Delta \theta = \omega_i t + \frac{1}{2}\alpha t^2$ [7.8]
$v^2 = v_i^2 + 2a\Delta x$	$\omega^2 = \omega_i^2 + 2\alpha\Delta \theta$ [7.9]

Pure Translation (Fixed Direction)		Pure Rotation (Fixed Axis)	
Position	x	Angular position	θ
Velocity	$v = dx/dt$	Angular velocity	$\omega = d\theta/dt$
Acceleration	$a = dv/dt$	Angular acceleration	$\alpha = d\omega/dt$
Mass	m	Rotational inertia	I
Newton's second law	$F_{\text{net}} = ma$	Newton's second law	$\tau_{\text{net}} = I\alpha$
Work	$W = \int F dx$	Work	$W = \int \tau d\theta$
Kinetic energy	$K = \frac{1}{2}mv^2$	Kinetic energy	$K = \frac{1}{2}I\omega^2$
Power (constant force)	$P = Fv$	Power (constant torque)	$P = \tau\omega$
Work-kinetic energy theorem	$W = \Delta K$	Work-kinetic energy theorem	$W = \Delta K$

relationship between angular and linear quantities:

- Displacements: $s = \theta r$
- Speeds: $v_t = \omega r$
- Acceleration: $a_t = \alpha r$

For a rotating object, it has the same angular motion but not have the same linear motion.

Centripetal Acceleration

The acceleration is directed **toward the center** of the circle of motion. The magnitude of the centripetal acceleration is

$$a_c = \frac{v^2}{r} = \omega^2 r$$

And Newton's Second Law says the centripetal acceleration is accompanied by a force.

$$F_c = ma_c$$

The force stands for **any force** keeps an object following a circular path. it is a **fictitious force**.

Total Acceleration

The tangential components of the acceleration is due to changing speed. The centripetal component of the acceleration is due to changing direction. And total acceleration is

$$a = \sqrt{a_t^2 + a_c^2}$$

Vector Nature of Angular Quantities

Angular displacement, velocity and acceleration are all **vector quantities**. But the direction is defined by using **right hand rule**.

Newton's Law of Universal Gravitation

The force between any two particles.

$$F = G \frac{m_1 m_2}{r^2}$$

It is an example of an **inverse square law**. G is a constant of universal gravitational, determined experimentally by Henry Cavendish.

Gravitational Potential Energy

For object **high above** the earth's surface, the gravitational potential energy is

$$PE = -G \frac{M_E m}{r}$$

and zero reference level is infinitely far from the earth. $r = R_E + h$.

Near the earth's surface, we have

$$g = G \frac{M_E}{r^2}$$

and thus

$$PE = mgh$$

Cosmic Speed

- First cosmic velocity: The speed for an object to into the space. The least speed for a object orbits the planet.

$$G \frac{m m_E}{R^2} = \frac{m v_1^2}{R} = mg$$

Thus

$$v_1 = \sqrt{gR} = \sqrt{\frac{G m_E}{R}}$$

- Second cosmic velocity: The escape speed is the speed needed for an object to soar off into space and never return

$$\frac{1}{2} m v^2 - G \frac{m m_E}{r} = 0$$

thus

$$v = \sqrt{\frac{2Gm_E}{r}} = \sqrt{2gR}$$

Kepler's Laws

1. All planets move in **elliptical orbits** with the **Sun at one of the focal points**.
2. A line drawn from the Sun to any planet sweeps out **equal areas in equal time intervals**.
3. The square of the orbital period of any planet is proportional to cube of the average distance from the Sun to the planet. $T^2 = Kr^3$.

Communications Satellite

A geosynchronous orbit, remains above the same place on the earth with the period being 24hr.